

Supplementary Material

When AI Exposure Rankings Disagree: Robust Top-25 Selection Across 661 U.S. Occupations

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S1. Input and scenario register

The 18 primitive priority vectors are the Cartesian product of three exposure channels and six constructions. For occupation i and design t , priority is $p_i^t = s_i q_i^t$, where s_i is May 2025 OEWS employment share. The priority box is $\ell_i = \min_t p_i^t$, $u_i = \max_t p_i^t$; the point core is $c_i = p_i^{\text{AIOE, equal-five}}$, so $\ell_i \leq c_i \leq u_i$ for every occupation.

Table 1: Primitive scenario construction.

Axis	Admitted levels
Exposure channel	AIOE; Microsoft user-goal; Microsoft AI-action/nonphysical
Construction	equal-five; leave Critical Thinking out; leave Active Learning out; leave Learning Strategies out; leave Active Listening out; leave Speaking out

S2. Membership counts and the 14 stable occupations

Table 2: Membership counts by selection size.

k	Scenario intersection	Scenario union	Box guaranteed	Box possible
10	7	16	2	43
25	14	37	4	87
50	32	73	14	159
100	68	135	27	244

At $k = 25$, the 14 occupations present in every coherent Top-25 are:

1. General and Operations Managers;
2. Accountants and Auditors;
3. Registered Nurses;
4. Waiters and Waitresses;
5. Janitors and Cleaners, Except Maids and Housekeeping Cleaners;
6. Cashiers;
7. Retail Salespersons;
8. Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products;
9. Bookkeeping, Accounting, and Auditing Clerks;
10. Receptionists and Information Clerks;
11. Secretaries and Administrative Assistants, Except Legal, Medical, and Executive;
12. Office Clerks, General;

13. Heavy and Tractor-Trailer Truck Drivers;
14. Stockers and Order Fillers.

These are stable screening priorities within the declared 18-specification family; the result is not a displacement or program recommendation.

S3. Exact rank proof and tie convention

Best rank is existential:

$$r_i^{\min} = 1 + \sum_{j \neq i} \mathbf{1}\{\ell_j > u_i\}.$$

Worst rank is pessimistic:

$$r_i^{\max} = 1 + \sum_{j \neq i} \mathbf{1}\{u_j \geq \ell_i\}.$$

Possible membership is $r_i^{\min} \leq k$ and guaranteed membership is $r_i^{\max} \leq k$. The strict/non-strict distinction controls boundary inclusion in the contraction analysis.

S4. Safe-screening proof detail

Let $G = \{i : r_i^{\max} \leq k\}$ and $Z = \{i : r_i^{\min} > k\}$. To insert $g \in G$, a size- k list omitting it contains an h with $u_h < \ell_g$, and replacement weakly increases every scenario-wise selected mass. If $i \in Z$ is selected, at least k competitors satisfy $\ell_j > u_i$, so some omitted j strictly dominates i in every scenario. Each such exchange strictly increases every scenario-wise selected mass; because the family of size- k lists is finite, cycling is impossible. Therefore an optimum exists containing G and excluding Z for any minimization objective coordinatewise nonincreasing in scenario-wise selected mass.

S5. Fixed-list verification and criterion sensitivity

Table 3: Fixed-list comparison at $k = 25$ over 18 primitive scenarios.

Rule	Mean capture	Minimum capture	Max. rel. regret	Box regret
Scale only	93.02%	88.35%	11.65%	0.0238
Baseline composite	96.14%	93.70%	6.30%	0.0181
Relative minimax	96.99%	96.52%	3.48%	0.0195

The relative-minimax objective is $\max_t R_t(x)$ with $R_t(x) = 1 - \sum_i p_i^t x_i / V_t^*$. The absolute-minimax sensitivity instead defines

$$R_t^{\text{abs}}(x) = V_t^* - \sum_i p_i^t x_i, \quad \min_{x: \sum_i x_i = k} \max_t R_t^{\text{abs}}(x).$$

Its list overlaps the relative-minimax list in 23/25 occupations, with minimum relative capture 94.79% and maximum relative regret 5.21%.

S6. Nested-support sensitivity and exact boundary semantics

For retained width $g \in [0, 1]$,

$$[L_i(g), U_i(g)] = [c_i - g(c_i - \ell_i), c_i + g(u_i - c_i)].$$

These values are transition boundaries. Under the stated tie convention, the post-contraction state applies for $g < g^*$ when contraction proceeds downward from $g = 1$; equality is evaluated by the original strict $>$ and non-strict $\geq$ rules rather than by rounded prose labels.

At the 37-member boundary, the five possible-set-only occupations are Computer and Information Systems Managers (11-3021), Medical Assistants (31-9092), Billing and Posting Clerks (43-3021), First-Line Supervisors of Construction Trades and Extraction Workers (47-1011), and Inspectors, Testers, Sorters, Samplers, and Weighers (51-9061). The five

Table 4: Structural Top-25 transition boundaries.

Event	g^*	$1 - g^*$	Identity check
Guaranteed $\rightarrow$ 14	0.5743200867	0.4256799133	coherent-intersection set after crossing
Possible $\rightarrow$ 37	0.4346510223	0.5653489777	cardinality only; 32 shared
Full 25/25	0.0787564209	0.9212435791	point list

coherent-union-only occupations are Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031), Amusement and Recreation Attendants (39-3091), Counter and Rental Clerks (41-2021), Production, Planning, and Expediting Clerks (43-5061), and First-Line Supervisors of Production and Operating Workers (51-1011).

For a strictly decreasing homeomorphism $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 1$, $\phi(1) = 0$, set $g = \phi(\alpha)$. Linear $\phi(\alpha) = 1 - \alpha$ and quadratic $\phi(\alpha) = (1 - \alpha)^2$ relabel the same support family and cannot constitute independent robustness tests.

S7. Secondary SIPP diagnostic and claim boundary

A separate descriptive diagnostic compares the three Top-25 lists with a fixed 17-occupation target set. The scale-only list contains one target with outgoing SIPP support and captures 50.05% of target-origin weighted outgoing mass. Baseline composite and relative minimax each contain two supported targets and each capture 66.39%. These are descriptive overlaps only; they do not estimate out-of-sample regret, worker transition probability, AI causality, or program success.

Table 5: Permitted and unsupported interpretations.

Output	Permitted	Unsupported
Box membership	Exact membership under independent priority intervals	Probability of selection
Coherent intersection/union	Membership over 18 admitted designs	Universal robustness
Relative-minimax list	Best worst relative capture over declared scenarios	Optimal workforce policy
Box regret	Exact absolute regret under independent priority intervals	Dominance under coherent designs
Boundary g^*	Width at which rank-comparison status can change	Confidence or calibrated possibility
SIPP overlap	Descriptive coverage of external target set	Validation or transition probability